Assignment – 13.1

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Task 1 – Refactoring – Removing Code Duplication

PROMPT :

Legacy script with repeated logic

print("Area of Rectangle:", 5 * 10)

print("Perimeter of Rectangle:", 2 * (5 + 10))

print("Area of Rectangle:", 7 * 12)

print("Perimeter of Rectangle:", 2 * (7 + 12))

print("Area of Rectangle:", 10 * 15)

print("Perimeter of Rectangle:", 2 * (10 + 15))

#give o Refactored code with a reusable function and no duplication. and add docstrings to all functions.

CODE :

AI-13.1.py > ...

```python
# Legacy script with repeated logic
# print("Area of Rectangle:", 5 * 10)
# print("Perimeter of Rectangle:", 2 * (5 + 10))
# print("Area of Rectangle:", 7 * 12)
# print("Perimeter of Rectangle:", 2 * (7 + 12))
# print("Area of Rectangle:", 10 * 15)
# print("Perimeter of Rectangle:", 2 * (10 + 15))
#give o Refactored code with a reusable function and no duplication. and add docstrings to all functions.
def calculate_rectangle_properties(length, width):
    """Calculates and prints the area and perimeter of a rectangle given its length and width."""
    area = length * width  # Calculate area
    perimeter = 2 * (length + width)  # Calculate perimeter
    print(f"Area of Rectangle: {area}")  # Print area
    print(f"Perimeter of Rectangle: {perimeter}")  # Print perimeter
# Call the function with different rectangle dimensions
calculate_rectangle_properties(5, 10)
calculate_rectangle_properties(7, 12)
calculate_rectangle_properties(10, 15)
```

OUTPUT :

```
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
```

JUSTIFICATION :

The legacy code contained repeated calculations for rectangle area and perimeter, which violates the DRY (Don't Repeat Yourself) principle. Refactoring into a reusable function improves maintainability, readability, and scalability because any logic change needs to be updated only once.

Task 2 – Refactoring – Extracting Reusable Functions

PROMPT :

Legacy script with inline repeated logic

price = 250

tax = price * 0.18

total = price + tax

print("Total Price:", total)

price = 500

tax = price * 0.18

total = price + tax

print("Total Price:", total)

#give o Code with a function calculate_total(price) that can be reused for multiple price inputs.and add docstrings to all functions.

CODE :

```python
# Legacy script with inline repeated logic
# price = 250
# tax = price * 0.18
# total = price + tax
# print("Total Price:", total)
# price = 500
# tax = price * 0.18
# total = price + tax
# print("Total Price:", total)
#give o Code with a function calculate_total(price) that can be reused for multiple price inputs.and add docstrings to all functions.
def calculate_total(price):
    """Calculates and prints the total price including tax for a given price."""
    tax = price * 0.18  # Calculate tax
    total = price + tax  # Calculate total price
    print(f"Total Price: {total}")  # Print total price
# Call the function with different price inputs
calculate_total(250)
calculate_total(500)
```

OUTPUT :

```
Total Price: 295.0
Total Price: 590.0
```

JUSTIFICATION :

The tax calculation logic was repeated for different price values. Creating a reusable function calculate_total(price) modularizes the logic, reduces redundancy, and enables reuse for multiple inputs without rewriting code.

Task 3 – Refactoring Using Classes and Methods (Eliminating Redundant Conditional Logic)

PROMPT :

#marks = 85

if marks >= 90:

print("Grade A")

elif marks >= 75:

print("Grade B")

else:

print("Grade C")

marks = 72

if marks >= 90:

print("Grade A")

elif marks >= 75:

print("Grade B")

else:

print("Grade C")

#give a code with class named GradeCalculator with method calculate_grade(self,marks) inside the class and create object of the class and call the method with different marks to get the grade. print the returned grade values.

CODE :

```
#marks = 85
# if marks >= 90:
#     print("Grade A")
# elif marks >= 75:
#     print("Grade B")
# else:
#     print("Grade C")
# marks = 72
# if marks >= 90:
#     print("Grade A")
# elif marks >= 75:
#     print("Grade B")
# else:
#     print("Grade C")
#give a code with class named GradeCalculator with method calculate_grade(self,marks) inside the class and create object of the class and call the method with
class GradeCalculator:
    """A class to calculate and return the grade based on marks."""

    def calculate_grade(self, marks):
        """Calculates and returns the grade based on the given marks."""
        if marks >= 90:
            return "Grade A"
        elif marks >= 75:
            return "Grade B"
        else:
            return "Grade C"
# Create an object of the GradeCalculator class
grade_calculator = GradeCalculator()
# Call the calculate_grade method with different marks and print the results
print(grade_calculator.calculate_grade(85))  # Should print "Grade A"
print(grade_calculator.calculate_grade(72))  # Should print "Grade B"
print(grade_calculator.calculate_grade(60))  # Should print "Grade C"
```

OUTPUT :

```
Grade B
Grade C
Grade C
```

JUSTIFICATION :

Repeated conditional grading logic reduces maintainability and increases error risk. Implementing a class-based design encapsulates grading behavior, promotes reusability, and follows object-oriented principles such as abstraction and modularity.

Task 4 – Refactoring – Converting Procedural Code to Functions

PROMPT :

num = int(input("Enter number: "))

square = num * num

print("Square:", square)

#give Modular code using functions like get_input(), calculate_square(), and display_result().

CODE :

```python
# num = int(input("Enter number: "))
# square = num * num
# print("Square:", square)
#give Modular code using functions like get_input(), calculate_square(), and display_result().
def get_input():
    """Prompts the user to enter a number and returns it as an integer."""
    return int(input("Enter number: "))
def calculate_square(num):
    """Calculates and returns the square of the given number."""
    return num * num
def display_result(square):
    """Prints the square of the number."""
    print(f"Square: {square}")
# Main function to orchestrate the flow of the program
def main():
    num = get_input()  # Get user input
    square = calculate_square(num)  # Calculate the square of the number
    display_result(square)  # Display the result
# Call the main function to execute the program
if __name__ == "__main__":
    main()
```

OUTPUT :

```
Enter number: 6
Square: 36
```

JUSTIFICATION :

Separating input, processing, and output into individual functions enhances code clarity, supports testing of individual components, and follows functional decomposition for better maintainability.

Task 5 – Refactoring Procedural Code into OOP Design

PROMPT :

#salary = 50000

tax = salary * 0.2

net = salary - tax

print(net)

#give a code with A class like EmployeeSalaryCalculator with methods and attributes.

CODE :

```python
#salary = 50000
# tax = salary * 0.2
# net = salary - tax
# print(net)
#give a code with A class like EmployeeSalaryCalculator with methods and attributes.
class EmployeeSalaryCalculator:
    """A class to calculate the net salary of an employee after tax deduction."""

    def __init__(self, salary):
        """Initializes the EmployeeSalaryCalculator with the given salary."""
        self.salary = salary  # Store the salary as an attribute

    def calculate_tax(self):
        """Calculates and returns the tax based on the salary."""
        return self.salary * 0.2  # Calculate tax as 20% of the salary

    def calculate_net_salary(self):
        """Calculates and returns the net salary after deducting tax."""
        tax = self.calculate_tax()  # Get the calculated tax
        net_salary = self.salary - tax  # Calculate net salary
        return net_salary  # Return the net salary
# Create an object of the EmployeeSalaryCalculator class with a salary of 50000
employee_salary_calculator = EmployeeSalaryCalculator(50000)
# Calculate and print the net salary
net_salary = employee_salary_calculator.calculate_net_salary()  # Get the net salary
print(net_salary)  # Print the net salary
```

OUTPUT :

```
40000.0
```

JUSTIFICATION :

Encapsulating salary and tax computation within a class improves data hiding and organization. The OOP approach allows reuse, extension, and better representation of real-world entities like employees.

Task 6 – Optimizing Search Logic

PROMPT :

#users = ["admin", "guest", "editor", "viewer"]

name = input("Enter username: ")

found = False

for u in users:

if u == name:

found = True

print("Access Granted" if found else "Access Denied")

#give a code with Use of sets or dictionaries with complexity justification.

#Refactor inefficient linear searches using appropriate data structures.

CODE :

```
#users = ["admin", "guest", "editor", "viewer"]
# name = input("Enter username: ")
# found = False
# for u in users:
#     if u == name:
#         found = True
# print("Access Granted" if found else "Access Denied")
#give a code with Use of sets or dictionaries with complexity justification.
#Refactor inefficient linear searches using appropriate data structures.
users = {"admin", "guest", "editor", "viewer"}  # Use a set for O(1) average time complexity for lookups
name = input("Enter username: ")  # Get user input
if name in users:  # Check if the username is in the set
    print("Access Granted")  # Print access granted if found
else:
    print("Access Denied")  # Print access denied if not found
```

OUTPUT :

```
Enter username: viewer
Access Granted
```

JUSTIFICATION :

Linear search has O(n) complexity. Using a set or dictionary enables average O(1) lookup time, improving performance and demonstrating appropriate data structure selection.

Task 7 – Refactoring the Library Management System

PROMPT :

Library Management System (Unstructured Version)

This code needs refactoring into a proper module with documentation.

```
# # library_db = {}
# # # Adding first book
# # title = "Python Basics"
# # author = "John Doe"
# # isbn = "101"
# # if isbn not in library_db:
# #     library_db[isbn] = {"title": title, "author": author}
# #     print("Book added successfully.")
# # else:
# #     print("Book already exists.")
# # # Adding second book (duplicate logic)
# # title = "AI Fundamentals"
# # author = "Jane Smith"
# # isbn = "102"
# # if isbn not in library_db:
# #     library_db[isbn] = {"title": title, "author": author}
# #     print("Book added successfully.")
# # else:
# #     print("Book already exists.")
# # # Searching book (repeated logic structure)
# # isbn = "101"
# # if isbn in library_db:
# #     print("Book Found:", library_db[isbn])
# # else:
# #     print("Book not found.")
# # # Removing book (again repeated pattern)
# # isbn = "101"
# # if isbn in library_db:
```

del library_db[isbn]

print("Book removed successfully.")

else:

print("Book not found.")

Searching again

isbn = "101"

if isbn in library_db:

print("Book Found:", library_db[isbn])

else:

print("Book not found.")

#create a module library.py with functions o add_book(title, author, isbn) ,remove_book(isbn), search_book(isbn),generate document in the terminal,which can create html document, which can be able to open in a browser.

CODE :

```
# Library Management System (Unstructured Version)
# This code needs refactoring into a proper module with documentation.
# library_db = {}
# # Adding first book
# title = "Python Basics"
# author = "John Doe"
# isbn = "101"
# if isbn not in library_db:
#     library_db[isbn] = {"title": title, "author": author}
#     print("Book added successfully.")
# else:
#     print("Book already exists.")
# # Adding second book (duplicate logic)
# title = "AI Fundamentals"
# author = "Jane Smith"
# isbn = "102"
# if isbn not in library_db:
#     library_db[isbn] = {"title": title, "author": author}
#     print("Book added successfully.")
# else:
#     print("Book already exists.")
# # Searching book (repeated logic structure)
# isbn = "101"
# if isbn in library_db:
#     print("Book Found:", library_db[isbn])
# else:
#     print("Book not found.")
# # Removing book (again repeated pattern)
# isbn = "101"
# if isbn in library_db:
#     del library_db[isbn]
#     print("Book removed successfully.")
# else:
#     print("Book not found.")
# # Searching again
# isbn = "101"
# if isbn in library_db:
#     print("Book Found:", library_db[isbn])
# else:
#     print("Book not found.")
#create a module library.py with functions o    add_book(title, author, isbn) ,remove_book(isbn), search_book(isbn),generate document in the terminal,which can cr
# library.py
class Library:
    """A class to manage a library of books with functionalities to add, remove, and search for books."""

    def __init__(self):
        """Initializes the Library with an empty database."""
```

```python
#create a module library.py with functions o    add_book(title, author, isbn) ,remove_book(isbn), search_book(isbn),generate document in the terminal,which can create
# library.py
class Library:
    """A class to manage a library of books with functionalities to add, remove, and search for books."""

    def __init__(self):
        """Initializes the Library with an empty database."""
        self.library_db = {}  # Initialize an empty dictionary to store book information

    def add_book(self, title, author, isbn):
        """Adds a book to the library database if the ISBN is not already present."""
        if isbn not in self.library_db:
            self.library_db[isbn] = {"title": title, "author": author}  # Add book details to the database
            print("Book added successfully.")
        else:
            print("Book already exists.")

    def remove_book(self, isbn):
        """Removes a book from the library database based on the ISBN."""
        if isbn in self.library_db:
            del self.library_db[isbn]  # Remove the book from the database
            print("Book removed successfully.")
        else:
            print("Book not found.")

    def search_book(self, isbn):
        """Searches for a book in the library database based on the ISBN and returns its details."""
        if isbn in self.library_db:
            return self.library_db[isbn]  # Return book details if found
        else:
            return "Book not found."  # Return message if book is not found
# Example usage of the Library class
if __name__ == "__main__":
    library = Library()  # Create an instance of the Library class
    library.add_book("Python Basics", "John Doe", "101")  # Add a book
    library.add_book("AI Fundamentals", "Jane Smith", "102")  # Add another book
    print(library.search_book("101"))  # Search for a book
    library.remove_book("101")  # Remove a book
    print(library.search_book("101"))  # Search for the removed book
```

OUTPUT :

```
Book added successfully.
Book added successfully.
{'title': 'Python Basics', 'author': 'John Doe'}
Book removed successfully.
Book not found.
```

JUSTIFICATION :

The original script lacked modularity and repeated logic patterns. Creating reusable functions within a module improves code organization, enables documentation generation, and supports scalable system design.

Task 8– Fibonacci Generator

PROMPT :

fibonacci bad version

n=int(input("Enter limit: "))

a=0

b=1

print(a)

print(b)

for i in range(2,n):

c=a+b

print(c)

a=b

b=c

#refactor into a clean reusable function,add docstrings and test cases,compare refactored and original code.

CODE :

```python
# fibonacci bad version
# n=int(input("Enter limit: "))
# a=0
# b=1
# print(a)
# print(b)
# for i in range(2,n):
#  c=a+b
#  print(c)
#  a=b
#  b=c
#refactor into a clean reusable function,add docstrings and test cases,compare refactored and original code.
def fibonacci(n):
    """Generates and prints the Fibonacci sequence up to the nth term."""
    a, b = 0, 1  # Initialize the first two Fibonacci numbers
    for i in range(n):
        print(a)  # Print the current Fibonacci number
        a, b = b, a + b  # Update a and b to the next two Fibonacci numbers
# Test cases for the fibonacci function
if __name__ == "__main__":
    print("Fibonacci sequence up to 10 terms:")
    fibonacci(10)  # Test with n=10
    print("\nFibonacci sequence up to 15 terms:")
    fibonacci(15)  # Test with n=15
# Comparison of refactored and original code:
# The original code had a lot of repeated logic for printing the first two Fibonacci numbers and updating
# # the values of a and b. The refactored code uses a single loop to generate the Fibonacci sequence, eliminating redundancy and improving readability
```

OUTPUT :

```
Fibonacci sequence up to 15 terms:
0
1
1
2
3
5
8
13
21
34
55
89
144
233
377
```

JUSTIFICATION :

The initial implementation used global variables and procedural flow. Refactoring into a function enhances modularity, reusability, and testability while improving readability.

Task 9 – Twin Primes Checker

PROMPT :

twin primes bad version

a=11

b=13

fa=0

for i in range(2,a):

if a%i==0:

fa=1

fb=0

for i in range(2,b):

if b%i==0:

fb=1

if fa==0 and fb==0 and abs(a-b)==2:

print("Twin Primes")

else:

print("Not Twin Primes")

#refactor into is_prime(n) and is_twin_prime(p1,p2),add docstrings and generate a list of twin primes in a given range.

CODE :

```python
# twin primes bad version
# a=11
# b=13
# fa=0
# for i in range(2,a):
#  if a%i==0:
#   fa=1
# fb=0
# for i in range(2,b):
#  if b%i==0:
#   fb=1
# if fa==0 and fb==0 and abs(a-b)==2:
#  print("Twin Primes")
# else:
#  print("Not Twin Primes")
#refactor into is_prime(n) and is_twin_prime(p1,p2),add docstrings and generate a list of twin primes in a given range.
def is_prime(n):
    """Checks if a number n is prime."""
    if n <= 1:
        return False  # 0 and 1 are not prime numbers
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False  # n is divisible by a number other than 1 and itself
    return True  # n is prime
def is_twin_prime(p1, p2):
    """Checks if two numbers p1 and p2 are twin primes."""
    return is_prime(p1) and is_prime(p2) and abs(p1 - p2) == 2  # Check if both numbers are prime and differ by 2
def generate_twin_primes(start, end):
    """Generates a list of twin primes within a given range."""
    twin_primes = []
    for num in range(start, end):
        if is_twin_prime(num, num + 2):
            twin_primes.append((num, num + 2))  # Add the twin prime pair to the list
    return twin_primes  # Return the list of twin primes
# Example usage of the twin prime functions
if __name__ == "__main__":
    start_range = 10
    end_range = 100
    twin_primes_list = generate_twin_primes(start_range, end_range)  # Generate twin primes in the specified range
    print(f"Twin primes between {start_range} and {end_range}: {twin_primes_list}")  # Print the list of twin primes
```

OUTPUT :

```
Twin primes between 10 and 100: [(11, 13), (17, 19), (29, 31), (41, 43), (59, 61), (71, 73)]
```

JUSTIFICATION :

Inefficient prime checking and hardcoded values reduced flexibility. Creating reusable functions for prime and twin-prime checks improves efficiency, generalization, and maintainability.

Task 10 – Refactoring the Chinese Zodiac Program

PROMPT :

year = int(input("Enter a year: "))

if year % 12 == 0:

print("Monkey")

elif year % 12 == 1:

```
# #     print("Rooster")
# # elif year % 12 == 2:
# #     print("Dog")
# # elif year % 12 == 3:
# #     print("Pig")
# # elif year % 12 == 4:
# #     print("Rat")
# # elif year % 12 == 5:
# #     print("Ox")
# # elif year % 12 == 6:
# #     print("Tiger")
# # elif year % 12 == 7:
# #     print("Rabbit")
# # elif year % 12 == 8:
# #     print("Dragon")
# # elif year % 12 == 9:
# #     print("Snake")
# # elif year % 12 == 10:
# #     print("Horse")
# # elif year % 12 == 11:
# #     print("Goat")
# #give a reusable function get_zodiac(year) replace if-elif with a list or dictionary,add docstrings.
```

CODE :

```python
# year = int(input("Enter a year: "))
# if year % 12 == 0:
#     print("Monkey")
# elif year % 12 == 1:
#     print("Rooster")
# elif year % 12 == 2:
#     print("Dog")
# elif year % 12 == 3:
#     print("Pig")
# elif year % 12 == 4:
#     print("Rat")
# elif year % 12 == 5:
#     print("Ox")
# elif year % 12 == 6:
#     print("Tiger")
# elif year % 12 == 7:
#     print("Rabbit")
# elif year % 12 == 8:
#     print("Dragon")
# elif year % 12 == 9:
#     print("Snake")
# elif year % 12 == 10:
#     print("Horse")
# elif year % 12 == 11:
#     print("Goat")
#give a reusable function get_zodiac(year) replace if-elif with a list or dictionary,add docstrings.
def get_zodiac(year):
    """Returns the Chinese zodiac sign for a given year."""
    zodiac_signs = [
        "Monkey", "Rooster", "Dog", "Pig", "Rat", "Ox",
        "Tiger", "Rabbit", "Dragon", "Snake", "Horse", "Goat"
    ]  # List of zodiac signs in order
    return zodiac_signs[year % 12]  # Return the zodiac sign based on the year modulo 12
# Example usage of the get_zodiac function
if __name__ == "__main__":
    year_input = int(input("Enter a year: "))  # Get user input for the year
    zodiac_sign = get_zodiac(year_input)  # Get the zodiac sign for the input year
    print(f"The Chinese zodiac sign for the year {year_input} is: {zodiac_sign}")  # Print the zodiac sign

```

OUTPUT :

```
Enter a year: 6
The Chinese zodiac sign for the year 6 is: Tiger
```

JUSTIFICATION :

A long if–elif chain reduces readability and increases maintenance complexity. Using a list or dictionary simplifies mapping logic, improves clarity, and supports easier updates.

Task 11 – Refactoring the Harshad (Niven) Number Checker

PROMPT :

Harshad Number Checker (Unstructured Version)

num = int(input("Enter a number: "))

temp = num

sum_digits = 0

while temp > 0:

digit = temp % 10

sum_digits = sum_digits + digit

temp = temp // 10

if sum_digits != 0:

if num % sum_digits == 0:

print("True")

else:

print("False")

else:

print("False")

#create a resuable function is_harshad(number),accepts an integer,return true if the number is divisible by sum of its digits,return false,add docstrings.

CODE :

```
# Harshad Number Checker (Unstructured Version)
# num = int(input("Enter a number: "))
# temp = num
# sum_digits = 0
# while temp > 0:
#     digit = temp % 10
#     sum_digits = sum_digits + digit
#     temp = temp // 10
# if sum_digits != 0:
#     if num % sum_digits == 0:
#         print("True")
#     else:
#         print("False")
# else:
#     print("False")
#create a resuable function is_harshad(number),accepts an integer,return true if the number is divisible by sum of its digits,return false,add docstrings.
def is_harshad(number):
    """Checks if a number is a Harshad number (divisible by the sum of its digits)."""
    temp = number  # Store the original number in a temporary variable
    sum_digits = 0  # Initialize the sum of digits to 0
    while temp > 0:
        digit = temp % 10  # Get the last digit
        sum_digits += digit  # Add the digit to the sum
        temp //= 10  # Remove the last digit
    if sum_digits != 0:  # Check if the sum of digits is not zero to avoid division by zero
        return number % sum_digits == 0  # Return True if number is divisible by sum of digits, else False
    return False  # Return False if sum of digits is zero
# Example usage of the is_harshad function
if __name__ == "__main__":
    number_input = int(input("Enter a number: "))  # Get user input for the number
    result = is_harshad(number_input)  # Check if the number is a Harshad number
    print(result)  # Print the result (True or False)
```

OUTPUT :

```
Enter a number: 10
True
PS D:\AI>
```

JUSTIFICATION :

Mixing input handling with logic and using print-based output limits reuse. Refactoring into a boolean-returning function separates concerns, improves modularity, and supports testing.

Task 12 – Refactoring the Factorial Trailing Zeros Program

PROMPT :

Factorial Trailing Zeros (Unstructured Version)

n = int(input("Enter a number: "))

fact = 1

i = 1

while i <= n:

fact = fact * i

i = i + 1

count = 0

while fact % 10 == 0:

count = count + 1

fact = fact // 10

print("Trailing zeros:", count)

#create a reusable function count_trailing_zeros(n),accepts a non-negative integer,return the number of trailing zeros in n!,add docstrings.

CODE :

```python
# Factorial Trailing Zeros (Unstructured Version)
# n = int(input("Enter a number: "))
# fact = 1
# i = 1
# while i <= n:
#     fact = fact * i
#     i = i + 1
# count = 0
# while fact % 10 == 0:
#     count = count + 1
#     fact = fact // 10
# print("Trailing zeros:", count)
#create a reusable function count_trailing_zeros(n),accepts a non-negative integer,return the number of trailing zeros in n!,add docstrings.
def count_trailing_zeros(n):
    """Returns the number of trailing zeros in n! (factorial of n)."""
    count = 0  # Initialize count of trailing zeros to 0
    power_of_5 = 5  # Start with the first power of 5
    while power_of_5 <= n:  # Loop until the power of 5 is less than or equal to n
        count += n // power_of_5  # Add the number of multiples of the current power of 5
        power_of_5 *= 5  # Move to the next power of 5
    return count  # Return the total count of trailing zeros
# Example usage of the count_trailing_zeros function
if __name__ == "__main__":
    n_input = int(input("Enter a non-negative integer: "))  # Get user input for n
    trailing_zeros = count_trailing_zeros(n_input)  # Calculate the number of trailing zeros in n!
    print(f"Trailing zeros in {n_input}!: {trailing_zeros}")  # Print the result
```

OUTPUT :

```
Enter a non-negative integer: 5
Trailing zeros in 5!: 1
```

JUSTIFICATION :

Computing full factorial is inefficient for large inputs. Using the mathematical approach of counting multiples of 5 reduces time complexity significantly and improves performance.

Task 13 – Collatz Sequence Generator – Test Case Design

PROMPT :

#Collatz Sequence Generator – Test Case Design ,Test Cases to Design:,Normal: 6 → [6,3,10,5,16,8,4,2,1],Edge: 1 → [1],Negative: -5,Large: 27 (well-known long sequence),Requirement: Validate correctness with pytest.

CODE :

```python
#Collatz Sequence Generator – Test Case Design ,Test Cases to Design:,Normal: 6 → [6,3,10,5,16,8,4,2,1],Edge: 1 → [1],Negative: -5,Large: 27 (well-known long se
def collatz_sequence(n):
    """Generates the Collatz sequence for a given positive integer n."""
    if n <= 0:
        raise ValueError("Input must be a positive integer.")  # Validate input
    sequence = []  # Initialize an empty list to store the sequence
    while n != 1:  # Loop until n becomes 1
        sequence.append(n)  # Add the current value of n to the sequence
        if n % 2 == 0:  # If n is even
            n //= 2  # Divide n by 2
        else:  # If n is odd
            n = 3 * n + 1  # Multiply n by 3 and add 1
    sequence.append(1)  # Append the final value of 1 to the sequence
    return sequence  # Return the generated Collatz sequence
# Example usage of the collatz_sequence function
if __name__ == "__main__":
    print("Collatz sequence for 6:", collatz_sequence(6))  # Normal case
    print("Collatz sequence for 1:", collatz_sequence(1))  # Edge case
    try:
        print("Collatz sequence for -5:", collatz_sequence(-5))  # Negative case (should raise an error)
    except ValueError as e:
        print(e)  # Print the error message
    print("Collatz sequence for 27:", collatz_sequence(27))  # Large case
```

OUTPUT :

```
Collatz sequence for 1: [1]
Input must be a positive integer.
Collatz sequence for 27: [27, 82, 41, 124, 62, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, 242, 121, 364, 182, 91, 274, 137, 412, 206, 103, 310, 155, 466, 233, 700, 350, 175, 526, 263, 790, 39
 1186, 593, 1780, 890, 445, 1336, 668, 334, 167, 502, 251, 754, 377, 1132, 566, 283, 850, 425, 1276, 638, 319, 958, 479, 1438, 719, 2158, 1079, 3238, 1619, 4858, 2429, 7288, 3644, 1822, 911, 2734
 1367, 4102, 2051, 6154, 3077, 9232, 4616, 2308, 1154, 577, 1732, 866, 433, 1300, 650, 325, 976, 488, 244, 122, 61, 184, 92, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1]
```

JUSTIFICATION :

Designing normal, edge, negative, and large test cases ensures functional correctness, robustness, and reliability of the Collatz sequence implementation across input scenarios.

Task 14 – Lucas Number Sequence – Test Case Design

PROMPT : #Lucas Number Sequence – Test Case Design,Test Cases to Design:,Normal: 5 → [2, 1, 3, 4, 7],Edge: 1 → [2],Negative: -5 → Error,Large: 10 (last element = 76).,Requirement: Validate correctness with pytest.

CODE :

```python
#Lucas Number Sequence – Test Case Design,Test Cases to Design:,Normal: 5 → [2, 1, 3, 4, 7],Edge: 1 → [2],Negative: -5 → Error,Large: 10 (last element = 76).,Requir
def lucas_sequence(n):
    """Generates the Lucas number sequence up to the nth term."""
    if n <= 0:
        raise ValueError("Input must be a positive integer.")  # Validate input
    sequence = []  # Initialize an empty list to store the sequence
    for i in range(n):
        if i == 0:
            sequence.append(2)  # First term of Lucas sequence
        elif i == 1:
            sequence.append(1)  # Second term of Lucas sequence
        else:
            next_value = sequence[i - 1] + sequence[i - 2]  # Calculate the next term
            sequence.append(next_value)  # Append the next term to the sequence
    return sequence  # Return the generated Lucas sequence
# Example usage of the lucas_sequence function
if __name__ == "__main__":
    print("Lucas sequence for 5:", lucas_sequence(5))  # Normal case
    print("Lucas sequence for 1:", lucas_sequence(1))  # Edge case
    try:
        print("Lucas sequence for -5:", lucas_sequence(-5))  # Negative case (should raise an error)
    except ValueError as e:
        print(e)  # Print the error message
    print("Lucas sequence for 10:", lucas_sequence(10))  # Large case
```

OUTPUT :

```
Lucas sequence for 5: [2, 1, 3, 4, 7]
Lucas sequence for 1: [2]
Input must be a positive integer.
Lucas sequence for 10: [2, 1, 3, 4, 7, 11, 18, 29, 47, 76]
```

JUSTIFICATION :

Test case coverage validates sequence correctness, boundary handling, and error management, ensuring the function behaves reliably for different inputs.

Task 15 – Vowel & Consonant Counter – Test Case Design)

PROMPT : #Vowel & Consonant Counter – Test Case Design,Test Cases to Design:Normal: "hello" → (2,3),Edge: "" → (0,0),Only vowels: "aeiou" → (5,0).

CODE :

```python
#Vowel & Consonant Counter - Test Case Design,Test Cases to Design:Normal: "hello" → (2,3),Edge: "" → (0,0),Only vowels: "aeiou" → (5,0).
#Only consonants: "bcdfg" → (0,5),Requirement: Validate correctness with pytest.
def count_vowels_consonants(s):
    """Counts the number of vowels and consonants in a given string s."""
    vowels = "aeiouAEIOU"  # Define a string of vowels (both lowercase and uppercase)
    vowel_count = 0  # Initialize vowel count
    consonant_count = 0  # Initialize consonant count
    for char in s:  # Iterate through each character in the string
        if char.isalpha():  # Check if the character is an alphabet
            if char in vowels:  # Check if the character is a vowel
                vowel_count += 1  # Increment vowel count
            else:
                consonant_count += 1  # Increment consonant count
    return vowel_count, consonant_count  # Return the counts as a tuple
# Example usage of the count_vowels_consonants function
if __name__ == "__main__":
    print("Vowels and consonants in 'hello':", count_vowels_consonants("hello"))  # Normal case
    print("Vowels and consonants in '':", count_vowels_consonants(""))  # Edge case
    print("Vowels and consonants in 'aeiou':", count_vowels_consonants("aeiou"))  # Only vowels case
    print("Vowels and consonants in 'bcdfg':", count_vowels_consonants("bcdfg"))  # Only consonants case
```

OUTPUT :

```
Vowels and consonants in 'hello': (2, 3)
Vowels and consonants in '': (0, 0)
Vowels and consonants in 'aeiou': (5, 0)
Vowels and consonants in 'bcdfg': (0, 5)
```

JUSTIFICATION :

Multiple test cases including empty strings, vowel-only strings, and large inputs verify accuracy, edge-case handling, and performance of the counting logic.